Exchange surfaces

Why adaptations matter

1 Why exchange systems?

Single cells: **large SA:V;** their outer surface can meet exchange needs.

Large multicellular organisms: **lower SA:V,** long diffusion distances → specialised exchange surfaces and transport systems.

2 Surface area: volume



1 cm: 6:1 3 cm: 2:1

Cube side a: **SA = $6a^2$; V = a^3.**

1 cm cube: 6 ÷ 1 = **6:1.** 3 cm cube: 54 ÷ 27 = **2:1.**

Larger cube: more total area, less area per unit volume.

Cutting into pieces exposes more area without changing total volume.

3 Explain the adaptation

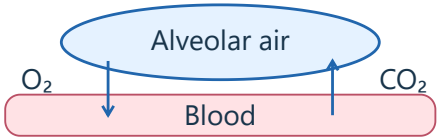
Large area: more particles cross at once. **Thin barrier:** shorter diffusion path.

Blood supply: brings/removes substances to maintain gradients. **Ventilation:** refreshes air/water.

Link **adaptation → effect → faster exchange.**

Animal exchange

4 Lung alveoli



O_2: air → blood; CO_2: reverse

Many tiny air sacs → large area; thin alveolar/capillary walls → short path.

Blood supply and ventilation maintain steep gradients. **O_2: air → blood. CO_2: blood → air.** An alveolus is one air sac, not a lung.

5 Small intestine

Villi/microvilli: large area. Thin lining: short diffusion distance.

Blood removes absorbed substances, maintaining gradients. Sugars can enter blood by **active transport** against a gradient.

6 Fish gills

Many **filaments/lamellae** → large area; thin surfaces → short diffusion distance.

Blood carries substances to/from the surface; flowing water maintains gradients. **O_2: water → blood; CO_2 moves out.**

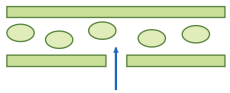
Plant exchange

7 Plant roots

Root hairs: large contact area with soil. Water enters by **osmosis** through a partially permeable membrane.

Minerals can enter against the gradient by **active transport,** using respiration energy.

8 Plant leaves



CO_2 enters through a stoma

Broad → large area. Thin → short diffusion distance. **Stomata** let gases in/out; internal **air spaces** shorten paths.

For photosynthesis: CO_2 enters, O_2 can diffuse out. State substance and direction in context.